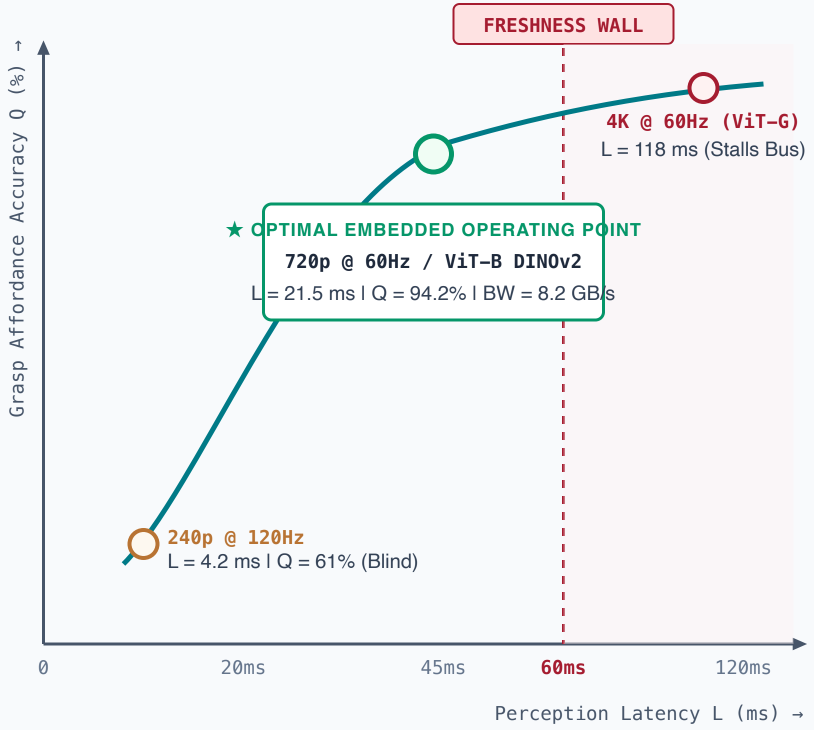
THE MULTI-OBJECTIVE PERCEPTION OPERATING FRONTIER

Navigating the five-variable Pareto trade-off: Quality (Q), Latency (L), Memory (M), Energy (E), and Bus Bandwidth (B)

PARETO OPTIMAL TRADEOFF: ACCURACY VS. SENSE-TO-ACTUATION LATENCY



5-VARIABLE EMBEDDED SYSTEM BUDGET AUDIT

❌ CLOUD ML MINDSET: 4K @ 60Hz + ViT-GIANT (7B VLA)

- **Quality (Q):** 98.1% (Superfluous high resolution)
- **Latency (L):** 118 ms (Freshness Wall breached, P99 = 240 ms)
- **Bus Bandwidth (B):** 31.4 GB/s (92% saturation of LPDDR5)
- **Power & Thermal (E):** 38 Watts (Rapid thermal throttle to 5 Hz)

✓ PHYSICAL AI SWEET SPOT: 720p @ 60Hz + DINOv2 / ViT-B

- **Quality (Q):** 94.2% (Millimeter spatial accuracy for grasping)
- **Latency (L):** 21.5 ms (Bounded P99 ≤ 24.2 ms with zero-copy DMA)
- **Bus Bandwidth (B):** 8.2 GB/s (Leaves 75% headroom for CPU/MCU)
- **Power & Thermal (E):** 12.5 Watts (Sustained steady-state operation)

⚠️ TINYML RETROFIT: 240p @ 120Hz + TINY CNN

- **Quality (Q):** 61.0% (Loses thin object edges & contact normal vectors)
- **Latency (L):** 4.2 ms (Extremely fast, but blind to depth nuances)
- **Bus Bandwidth (B):** 0.9 GB/s (Near-zero bus pressure)
- **Power & Thermal (E):** 4.1 Watts (Extremely low power)